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Design of a Predictive Digital Twin System for Large-Scale Varroa Management in Honeybee Apiaries

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Abstract

Varroa mites are a major global threat to honeybee colonies. Combining digital twins with scenario-generating models can be an enabler of precision apiculture, allowing for monitoring Varroa spread, generating treatment scenarios under varying conditions, and running remote interventions. This paper presents the conceptual design of this system for large-scale Varroa management in honeybee apiaries, with initial validation conducted through simulations and feasibility analysis. The design followed a design research framework. The proposed system integrates a wireless sensor network for continuous hive sensing, image capture, and remote actuation of treatment. It employs generative time-series models to forecast colony dynamics and a statistical network model to represent inter-colony spread; together, they support spread scenario prediction and what-if evaluations of treatments. The system evolves through continuous updates from field data, improving the accuracy of spread and treatment models over time. As part of our design research, an early feasibility assessment was carried out through the generation of synthetic data for spread model pretraining. In addition, a node-level energy budget for sensing, communication, and in-hive treatment was developed and matched with battery capacity and life calculations. Overall, this work outlines a path toward real-time, data-driven Varroa management across apiary networks, from regional to cross-border scales.

Keywords: digital twins; varroa mite mitigation; generative time-series models; wireless sensor networks; bee colony health monitoring; precision agriculture

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1. Introduction

The Varroa mite (Varroa destructor) presents a significant risk to honeybee colonies, leading to their decline and eventual collapse within a few years [1]. This issue affects not only beekeeping but also broader agricultural sectors reliant on bee pollination. Detected in Western Europe in the 1970s [2], the Varroa mite has since become a global concern. By June 2022, it had reached Australia, the last region previously unaffected, indicating its worldwide spread [3].

Various strategies are available for managing Varroa mites and are often combined within an integrated pest management (IPM) framework. These strategies can be classified as cultural, mechanical, biological, or chemical [4]. Cultural controls involve breeding resistant bee lines selected for their genetic capacity to reduce mite populations, though achieving optimal effectiveness with this approach may take decades. Mechanical controls include techniques such as hyperthermia, screened bottom boards, and drone brood

trapping, all of which physically remove mites from colonies. Biological controls employ parasitic fungi [5,6] and natural predators like pseudoscorpions [7] to organically reduce mite levels. Lastly, chemical methods utilize natural treatments, such as formic and oxalic acids, and synthetic treatments, like Amitraz, which offer more aggressive mite control [8].

Each method varies in terms of risk, efficacy, and response time. Cultural approaches typically pose the lowest risks and may be more sustainable in the long term, but their effectiveness is currently limited and needs improvement. Chemical methods are highly effective but are associated with higher risks, including the potential for residue accumulation in wax and honey, environmental hazards [2], and the development of resistant mite populations. The hyperthermia method, a mechanical approach, provides a balanced option, offering moderate effectiveness with manageable risks. Thermal treatments exploit the tolerance difference between bees (45 °C) and mites (41 °C), aiming to eliminate mites without harming bees, yet precise temperature control remains challenging [2].

A need-based strategy of closely monitoring the infection ratio and responding accordingly is recommended [2]. An implementation of this approach, called traffic light warning system, categorizes infestation rates into three thresholds (red, orange, or green) tailored to specific regions and times, guiding appropriate actions [2]. However, traditional infestation assessments are labor-intensive, requiring manual sampling of about 300 worker bees to detect mites [9,10]; this stresses bees and potentially misses early infestation stages [11].

The above-mentioned inefficiency has motivated efforts towards automation and precision, mainly through image or video analysis or other sensor technologies. Examples of visual approaches include sticky board imaging, where mites are automatically counted from photographs taken by smartphones or portable scanners [12–14]. Another study explores hyperspectral imaging to distinguish mites from bees based on spectral signatures [15]. Moreover, a set of other studies document the direct detection of mites on adult bees using typical image-detection techniques [11,16–19]. In one other example, short videos are fed to a model based on Vision Transformer (ViT) to detect mites [20]. These visual methods benefit from relatively straightforward interpretation and the possibility of being integrated into existing beekeeping routines, but they might require consistent imaging conditions or specialized hardware. The availability of commercial services and products based on visual approaches [21] can serve as an indicator of the feasibility of these methods.

Non-visual approaches aim to detect mite presence indirectly through their impact on the hive environment. One line of work employs vibrational sensors placed in brood combs, where mites produce characteristic signals that can be separated from normal colony vibrations using machine learning methods [22]. In another example, an electronic-nose system with metal-oxide sensors monitors volatile organic compounds in beehive air, capturing odor signatures from bees, mites, and hive microbiota to flag Varroa infestations non-invasively [23]. Together, these developments illustrate a trend toward both visual and non-visual strategies that can support automated Varroa surveillance as part of precision apiculture systems.

Incorporating these detection methods into a wireless sensor network (WSN) can streamline beehive health monitoring, reducing manual inspection errors and labor [24–26]. A WSN offers a robust framework for continuous health surveillance in beehives.

From another angle, modeling disease spread and population impacts across colonies provides a comprehensive understanding of infestation dynamics and underpins effective management strategies. Machine learning and mathematical approaches allow for the analysis of mite reproduction, bee population dynamics, and the environmental drivers of Varroa spread. For example, a mathematical model captures the coupled spread of Varroa

mites and associated diseases within colonies [27]. Other models offer insight into colony behavior and dynamics under Varroa pressure [28,29].

Exploring the spread of Varroa mites also involves analyzing how infestations move between colonies. Spatial network analysis offers a method to model these interactions, focusing on the pathways through which Varroa mites invade and affect bee populations [30]. Such models are instrumental in identifying effective control measures by understanding the spatial dynamics of disease spread.

Digital twins sit at the intersection of predictive modeling and sensor networking for effective Varroa management. A digital twin is a virtual model that mirrors a physical entity, combining various data inputs for processing and ensuring a two-way data exchange between the virtual and physical realms. It requires synchronization to accurately reflect changes in a physical object's condition or the actuating processes or mechanisms within it [31]. Digital twins transcend the notion of being mere digital replicas; they can be augmented through the integration of simulations and predictive models [32].

There is a growing trend in the application of digital twins in agriculture [33–35]. A recent bibliometric analysis of research on the use of digital twin technology in smart agriculture [34] highlights that most work currently focuses on crop monitoring, predictive modeling, disaster prediction, and simulation-based decision support. In another study [36], the authors examined how agricultural digital twins fuse sensor data, satellite imagery, and crop and soil models to support functions such as growth prediction, yield estimation, and precision resource management. They analyzed concrete use cases spanning irrigation, fertilization, pest control, and harvest planning,

Digital twin applications in agriculture are increasingly being extended to beekeeping [37]. One example is HONEYBEE-pDT [38], a prototype that integrates the BEEHAVE colony model [39] with large-scale geospatial datasets to simulate honeybee dynamics under varying landscapes and climate conditions. Unlike continuously updated twins, this system currently operates on offline data, though future plans include integrating hive monitoring inputs such as colony weight [38]. While not explicitly labeled as digital twins, other studies have adopted similar concepts and architectures. For instance, Senger et al. [40] presented a sensor-and-model-based system that compares predicted and actual hive weight changes to detect anomalies, enabling early alerts for abnormal colony developments and supporting timely interventions. Similarly, Johannsen et al. [41] focused on urban beekeeping, developing a multi-agent model that combines hive sensor data, beekeeper records (via the BEEP app), and weather data to simulate colony and management dynamics. This approach provides decision support for beekeepers (e.g., feeding, treatment, and swarming) and allows policymakers to explore regulatory impacts on urban beekeeping. Finally, a feasibility study [42], supervised by the authors, demonstrated a digital twin-based Varroa control system. The work showed reliable temperature regulation and communication while emphasizing affordability through low-cost sensors, simple thermal control, and the use of the open-source Eclipse Ditto platform.

This study contributes to ongoing advancements by presenting the conceptual design of a predictive digital twin for large-scale Varroa management. The system integrates a WSN for real-time data acquisition with generative scenario modeling to predict colony dynamics and simulate alternative treatment pathways, including hyperthermia. Models are dynamically updated as new data become available, enabling forecasts of population trends and supporting adaptive treatment decisions. Guided by the design research methodology (DRM) [43], the overarching research question is the following: How can a predictive digital twin be designed to support adaptive and scalable Varroa control in honeybee colonies? The aim is to propose and validate this conceptual framework, with early validation provided through simulation and feasibility analysis.

2. Method

Since this study is grounded in a design research approach, we employed the DRM framework, as outlined in [43]. Our work followed its structured progression through the stages of Research Clarification, Descriptive Study I, and Prescriptive Study and was extended partially into Descriptive Study II. In this regard, the Research Clarification stage is mapped mostly to the introduction part, where it defines the Varroa mite problem, explains why traditional methods are insufficient, and positions digital twins enhanced with generative sequence models for what-if scenario modeling as a potential solution.

Next, in the Descriptive Study I stage, we reviewed and summarized current approaches to the visual detection of Varroa mites, including You Only Look Once (YOLO), Faster Region-based Convolutional Neural Network (R-CNN), and ViT. In addition, we examined existing mathematical and computational models for simulating vertical (within-colony) and horizontal (between-colony) infection dynamics, such as differential equation models, the BEEHAVE framework, and the exponential random graph model (ERGM).

Next, in the Prescriptive Study stage, we outlined the architecture and design components of the proposed digital twin system, including its sensor networks, predictive models, and treatment actuators. We then proposed strategies for pretraining models of population dynamics and mite spread, describing how these models can be constructed and trained. For horizontal (inter-colony) spread, we employed an ERGM, whereas for vertical (intra-colony) dynamics, we used differential equation models, as described by Torres and Torres [29], to generate the required synthetic training data.

We conducted an early validation of the proposed system at the Descriptive Study II stage, using illustrative simulations of horizontal Varroa mite spread under both treatment and non-treatment scenarios. In this regard, an ERGM model was implemented to demonstrate the feasibility of generating synthetic data for training the digital twin models prior to the availability of large-scale field data. The model was approximated on geographic coordinates of bee apiaries in southern Sweden [44] to simulate regional Varroa transmission. The distribution statistics are provided in Table 1.

The ERGM simulation (using the EpiModel 2.4.0 library in R 4.5.1) was parameterized with 620 approximate beehive locations over a three-year period and executed in both susceptible, infected (SI) and susceptible, infected, recovered (SIR) modes to represent no intervention and intervention conditions. The network-formation model was based on the number of edges, with each hive connected to an average of four others, preferentially linking geographically proximate colonies. This design reflects realistic yet limited contact opportunities, such as drifting, robbing, swarming, beekeeper operations, and hive transport.

Infection probability was modeled as a uniform, seasonally varying rate (0.015 during non-winter months and 0.001 during winter), with each active hive-to-hive connection assumed to allow an average of 0.20 transmission opportunities per day. Recovery in the SIR mode was modeled as management-induced clearance (e.g., hyperthermia), with a daily probability of $\frac{1}{120}$ ($\approx$0.83%), corresponding to a mean infectious period of 120 days. To assess robustness, we further tested ten alternative intervention durations ranging from 30 to 365 days, in addition to the no recovery (SI) baseline.

Additionally, an energy consumption and feasibility analysis of the sensors and heaters was carried out based on available components.

Going further, we used the previous components and approach within the framework of adaptive surveillance strategies for Varroa management, describing how the dynamic monitoring strategy should be formed.

Table 1. Bee colony node distribution details.

Parameter	Value
Total Nodes	619
Average Distance Between Nodes	243.90 km
Distance Variability (SD)	122.85 km
Maximum Distance Between Nodes	658.68 km
Minimum Distance Between Nodes	0 km

3. Results

As the main outcome of the Prescriptive Study stage within the DRM framework, this work presents the conceptual design of a digital twin for Varroa management. The design encompasses the overall architecture, the specification of key components, an analysis of suitable Varroa mite detection approaches, and the proposal of scenario-generation models based on generative time-series techniques. In addition, with regard to the Descriptive Study II phase of the DRM framework, the results include an early validation step through the creation of synthetic pretraining data, as well as a feasibility assessment via node-level energy budget analysis.

3.1. Architecture and Components

Figure 1 depicts the main components of our digital twin design, and Figure 2 shows the main workflow. As illustrated in Figure 1, the digital twin system consists of the physical beehives (Figure 1: Ⓐ) and their corresponding digital twin counterparts (Figure 1: Ⓑ). Some of the beehives are equipped with sensors, and additional methods are employed to monitor environmental conditions (Figure 1: Ⓒ). The digital twin instances are integrated with models for predicting infection spread, running what-if simulations, and recommending treatments (Figure 1: Ⓔ). Furthermore, the system includes components for implementing planned interventions and treatments (Figure 1: Ⓓ).

Finally, a wireless sensor network facilitates communication between the components (Figure 1: Ⓕ). The workflow of our digital twin system is structured as a loop, indicated by the flow of thick blue lines. The current status of the beehives, particularly infection levels, is used to predict potential future scenarios and assess responses to various treatment strategies, as illustrated in Figure 2.

In this workflow (Figure 2), the infection status of beehives is continuously monitored by sampling data through a WSN (①). The digital twin instances are then updated with the latest beehive conditions (②). These updates are derived from either direct sensor readings, extrapolation of sampled data, or models, with parameters adjusted based on the most recent sensor data.

Based on the updated status, a prediction about the progression of Varroa mite infection is generated (④). To address these predicted scenarios, a response must be developed, taking into account environmental and infrastructural constraints, as well as overarching agricultural and environmental policies. To formulate this response, various feasible target scenarios are envisioned, each reflecting different trade-offs between capacity and environmental considerations (⑤–⑤$_n$).

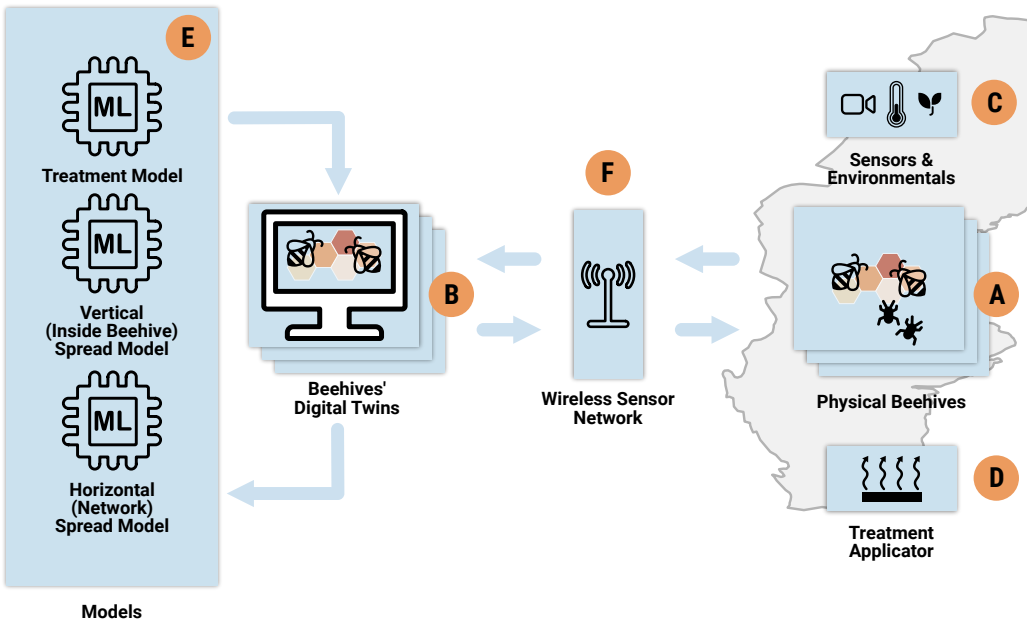


Figure 1. Main components of the digital twin system.

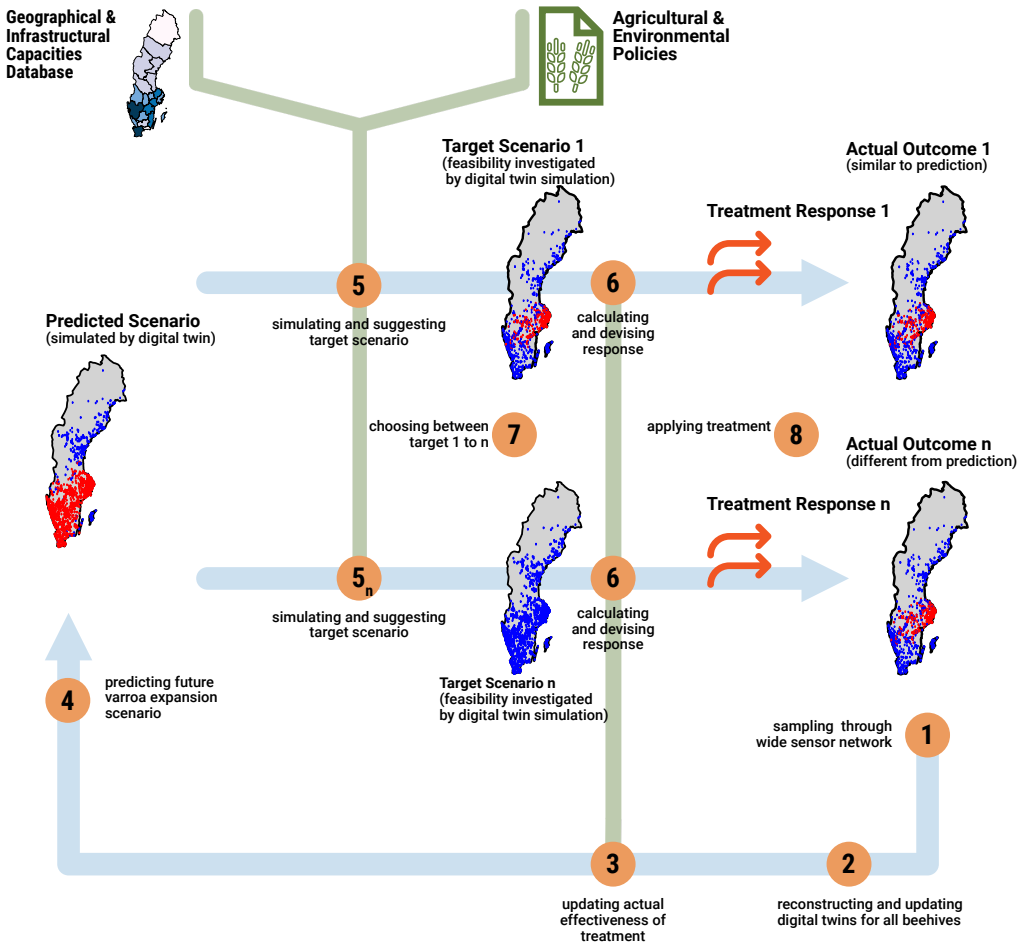


Figure 2. Main workflow of the digital twin system.

Each envisioned scenario results in a response characterized by varying levels of cost, impact, and feasibility (⑥–⑥$_n$). After the initial cycle in the workflow, the effectiveness of

prior interventions—as reflected in the actual outcomes—becomes an additional parameter (⑥) that is updated alongside other parameters (③).

Finally, based on policy considerations, managerial decisions, and the assessed cost, impact, and feasibility of each scenario, one is selected as the optimal choice (⑦). The chosen intervention is then implemented (⑧), often through remote commands to applicators in the beehives to perform specific actions, such as adjusting the temperature for hyperthermia treatment. Intervening with or the natural progression of the situation will result in new conditions, which will be sensed and fed back into the system as part of the loop (①).

3.2. Mite Detection System

The detection of Varroa mites in a beehive and measuring their level of spread are crucial for the design of the digital twin system. The visual detection of Varroa mites utilizes a series of techniques [45,46], ranging from conventional object detection [11,45] to hyperspectral imagery [15]. The actual source locations of the images are brood cells, sticky boards, the hive entrance, and hive frames [47].

A wide range of object-detection methods is available, and new variants appear rapidly. For the purpose of detecting Varroa mites, studies have applied YOLO and Faster R-CNN [11,47,48]. Recent work also positions ViT models as competitive in terms of accuracy for image-based tasks [49], although practical deployment on constrained devices requires care [47,50]. This gap is narrowing as edge-efficient Transformer designs continue to improve [51]. Accordingly, we considered ViT-based detectors within our digital twin framework while remaining model-agnostic pending in situ testing. The attention mechanism in Transformer-based models may further help focus on salient regions and support interpretability in hive images. Overall, this integration strengthens the digital twin's ability to mirror real-world conditions using state-of-the-art visual detection while allowing for flexibility to adopt newer or lighter models as evidence accumulates.

Building on this foundation, recent studies further demonstrate the potential of Vision Transformer (ViT) models for automated Varroa detection. Giovannesi et al. [20] presented VIT4V, a member of the ViT family, as a video-based classification system that leverages temporal sequences to detect Varroa mites. The method achieves strong results with 0.986 accuracy and 0.988 F1-score on full-video evaluations, processing 32-frame clips in approximately 250 ms using around 800 MB of memory on an Nvidia GeForce 1080 GPU. The system demonstrates robustness under practical conditions, such as blur, occlusions, dirt on hive plexiglass, and variable lighting, and is supported by the Varroa Destructor Video dataset (VD2), consisting of over 600 videos with RGB and thermal imaging [20]. In parallel, reports on still-image classification using ViT models also indicate high accuracy [52], though the setup of Giovannesi et al. [20] more closely reflects the realistic conditions of deploying a sensor package within a digital twin design.

3.3. Modeling Vertical and Horizontal Dynamics

The spread of mites within a single bee colony, referred to as vertical spread, is often studied using population models. In contrast, the spread between colonies, or horizontal spread, is typically examined using network-based models. Both processes can be investigated through mathematical or computational approaches [53]. Mathematical models, particularly those based on differential equations, offer a structured framework to describe disease transmission using parameters such as infection, recovery, and mortality rates [54,55]. Computational approaches, on the other hand, frequently rely on simulations—such as agent-based, stochastic, or network models—to capture the more complex dynamics that emerge from individual behaviors and spatial interactions [56].

In both cases, the complexity of biological processes is simplified into a smaller set of essential parameters, such as survival rates, transmission rates, and grooming rates, along with their interactions. This reductionist approach has its challenges, as it may not fully reflect the complexities of real-world conditions. By reducing processes to a small set of parameters or clear-cut interactions, these models may overlook relevant variables and the natural variability that occurs in actual settings, leading to large discrepancies between model predictions and actual outcomes.

Furthermore, traditional models often rely on coarse abstractions and are generally poorly suited for what-if analyses. Such analyses depend on many drivers beyond within-colony dynamics or disease transmission between colonies. Important drivers include but are not limited to beekeeping practices, monitoring and response routines, agricultural policies at regional and national levels, broader economic and industrial trends that shape land use and transportation, and both short-term variability and long-term trends in climate. While data and stand-alone prediction models may be available for many of these factors, integrating them into a single model of Varroa mite spread remains a significant challenge.

To address these challenges, we adopted two complementary directions. First, we developed digital twins that capture the real-time states of hives and other key entities. These twins continuously aggregate and update measurements, providing a consistent foundation for monitoring, visualization, forecasting, and what-if analysis. Second, we employed generative models for time series to project future trajectories and simulate counterfactual interventions. Such models learn from historical data, avoid restrictive assumptions about factor interactions, and can uncover important dependencies that may be locally relevant or overlooked in previous analyses. They also enable the integration of diverse drivers, including biological, environmental, and operational factors, into a single predictive framework. Because of their generative nature, these models can be conditioned on alternative policies, practices, interventions, and timings, thus supporting rigorous what-if evaluation without the risks and costs of real-world trials.

A digital twin for each hive and other key entities (such as apiaries, treatment devices, or local climate inputs) provides a living, data-driven representation of the system. It continuously integrates sensor streams and field logs to maintain up-to-date estimates of colony conditions, including brood temperature, forager activity, and inferred mite load, with quantified uncertainty. This unified perspective supports high-resolution monitoring and facilitates the early detection of abnormal patterns. The twin also records interventions and contextual information, ensuring traceability and auditability of outcomes. When combined with generative models for time series, it can simulate alternative conditions and schedules, creating a practical workspace for what-if analysis and decision support. Moreover, the approach can be scaled to large numbers of hives with minimal additional effort, delivering consistent summaries, insights, and alerts that enable timely action by beekeepers and policymakers.

Several classes of generative models for time series offer features that are applicable to predicting and performing counterfactual analyses of beehive conditions. Among them, Transformers provide powerful sequence-modeling capabilities and are widely adopted in many domains, particularly for capturing complex dependencies across long sequences [57]. Their attention mechanisms allow for the modeling of distant relationships, which can be useful for connecting long-term colony dynamics with near-term hive states. However, this advantage is balanced by a high computational demand. At the same time, Transformer architectures typically operate on fixed-step inputs, unless modified through architectural variations [58,59]. This is a consideration for beehive data that may arrive at irregular intervals. These models have also become the foundation for large-scale systems, such as

large language models (LLMs), demonstrating their scalability and potential for adaptation across domains.

There are other model families, particularly state-space neural networks, which provide complementary features that align well with the characteristics of hive data. Examples include structured state-space sequence models, such as S4 [60] and Mamba [61], continuous-time approaches, like neural ordinary differential equation (N-ODE) [62], and latent-variable models, like deep Markov model (DMM) [63]. These approaches are particularly well-suited for time series with irregular sampling and multiple data sources, as hive measurements, environmental covariates, and management interventions often occur on different timelines. They also support long-horizon forecasts without requiring attention over very large windows, which makes them attractive for projecting colony health trajectories. Furthermore, these models include mechanisms for handling missing or sparse data through filtering, smoothing, and latent-state inference, an important consideration since only a subset of colonies may be continuously instrumented, while others rely on opportunistic measurements. These properties position state-space neural models as strong candidates for applications where both temporal and spatial gaps are present. Among state-space families, Deep Markov Models (DMMs) and related latent-variable approaches are inherently probabilistic, which makes them well-suited for "what-if" or counterfactual analysis; by explicitly modeling uncertainty over latent states and transitions, they can generate alternate trajectories under hypothetical interventions while quantifying confidence in those outcomes [63]. State-space models (SSMs), such as modern deep variants and Mamba, provide an efficient dynamical framework, where external control inputs can be injected to simulate interventions; however, when trained in a purely deterministic fashion, they lack the ability to represent uncertainty, limiting their use for principled counterfactual reasoning unless extended with probabilistic layers [64,65]. Transformers, in their standard form, are powerful sequence predictors but are similarly deterministic; probabilistic extensions (e.g., Variational Transformers [66])are emerging to bridge this gap by combining self-attention with generative uncertainty. Overall, the probabilistic nature of these models is what enables robust counterfactual analysis for our requirements regarding Varroa mite management scenario generation, while non-probabilistic versions can still support "what-if" simulations but require explicit probabilistic enhancements to move beyond black-box predictions. Finally, given the highly dynamic research landscape regarding these models, we aim to keep the range of options open.

To provide a more concrete understanding of DMM potential, we highlight the DMM architecture as an example. Figure 3 illustrates how a DMM can be trained for colony dynamics, while Figure 4 shows its application for forecasting and counterfactual analysis.

During the training phase, we have a set of observations describing beehive states, along with environmental and policy indicators. These are represented as vectors denoted by X, where X_t corresponds to the vector of all observations recorded at time or step t. The DMM model receives these observations as input (shown at the bottom of Figure 3) and produces outputs denoted by $\hat{X}$ (shown at the top of Figure 3). The model parameters are optimized so that the generated $\hat{X}_t$ series closely matches its corresponding X_t series. The U_t series represents intervention inputs, which may include intentional actions, such as Varroa mite treatments, or unintentional factors, such as weather changes. The vector B_0 encodes the baseline characteristics of each hive, which generally remain constant over time. The vector Z_t denotes the internal (latent) state of the model at time or step t. Although it is usually difficult to map the elements of Z_t directly to real-world indicators—as they are automatically learned—these latent variables can be interpreted as analogous to important features in traditional modeling, such as grooming rate, mite reproductive rate, or queen egg-laying rate. The sequence neural network that processes the X_t series is

a predictive architecture (e.g., recurrent neural network (RNN), gated recurrent unit (GRU), or long short-term memory (LSTM)) that learns temporal dependencies. However, these architectures alone do not naturally capture probabilistic or continuous transitions of the system's internal state, nor are they well-suited for what-if simulations. Therefore, only their hidden states h_t are used to construct Z_t. The transition neural network models the dynamics of changes in the latent state Z_t conditioned on interventions (U_t) and baseline characteristics (B_0), while the emission neural network is trained to translate latent states into observable outputs.

In prediction mode, we retained the transition and emission neural networks that were already trained to capture how the internal state of the system (the beehive alone or the beehive together with its surrounding environment) evolves over time under different interventions (U_t) and baseline characteristics (B_0). The emission network had already learned to translate these internal states into observable outputs about the beehives, which we denote as Y_t at each time step. For prediction tasks, we provided new sets of actual interventions (U_t) and baseline characteristics (B_0). For what-if (counterfactual) analyses, instead, we supplied alternative interventions and baseline characteristics. It should be noted that both the treatment model and the vertical spread model, depicted in component Ⓔ of Figure 1, were implemented within the same underlying architecture shown in Figures 3 and 4; they represent different uses of the same model by providing different intervention input data.

An important feature of this approach is that, within the digital twin system, training does not occur only once but is continuously repeated. This iterative process tunes the model to improve its predictive accuracy and to capture factors and dynamics that may have been previously overlooked. It corresponds to the feedback loop illustrated in Figure 1 between components Ⓑ and Ⓔ. Such a loop also helps address data drift and concept drift [67], where the underlying distributions and dynamics evolve over time and the model must be updated to reflect these changes.

When it comes to vertical spread modeling, there is a growing and promising body of research [68,69] that applies neural networks, particularly graph neural network (GNN), and more specifically, temporal-spatial graph neural network (TS-GNN). There are also approaches that employ frameworks such as ERGMs enhanced with neural network-based estimators of parameters or posteriors [70,71]. While these newer models may require time to mature and demonstrate consistent applicability, a comprehensive modeling of Varroa mite dynamics within each colony can already provide valuable inputs for more traditional ERGM-based approaches.

To summarize, digital twins of hives and other key entities, combined with generative state-space time-series models, provide a unified framework that integrates biological, environmental, and operational factors. This framework enables high-resolution, near-real-time monitoring of current colony conditions while also supporting forecasting and counterfactual what-if analysis of events, including different treatment decisions. This what-if analysis not only allows for the simulation of alternative scenarios but also assists in intervention scheduling and provides insights by quantifying uncertainty in the predicted results.

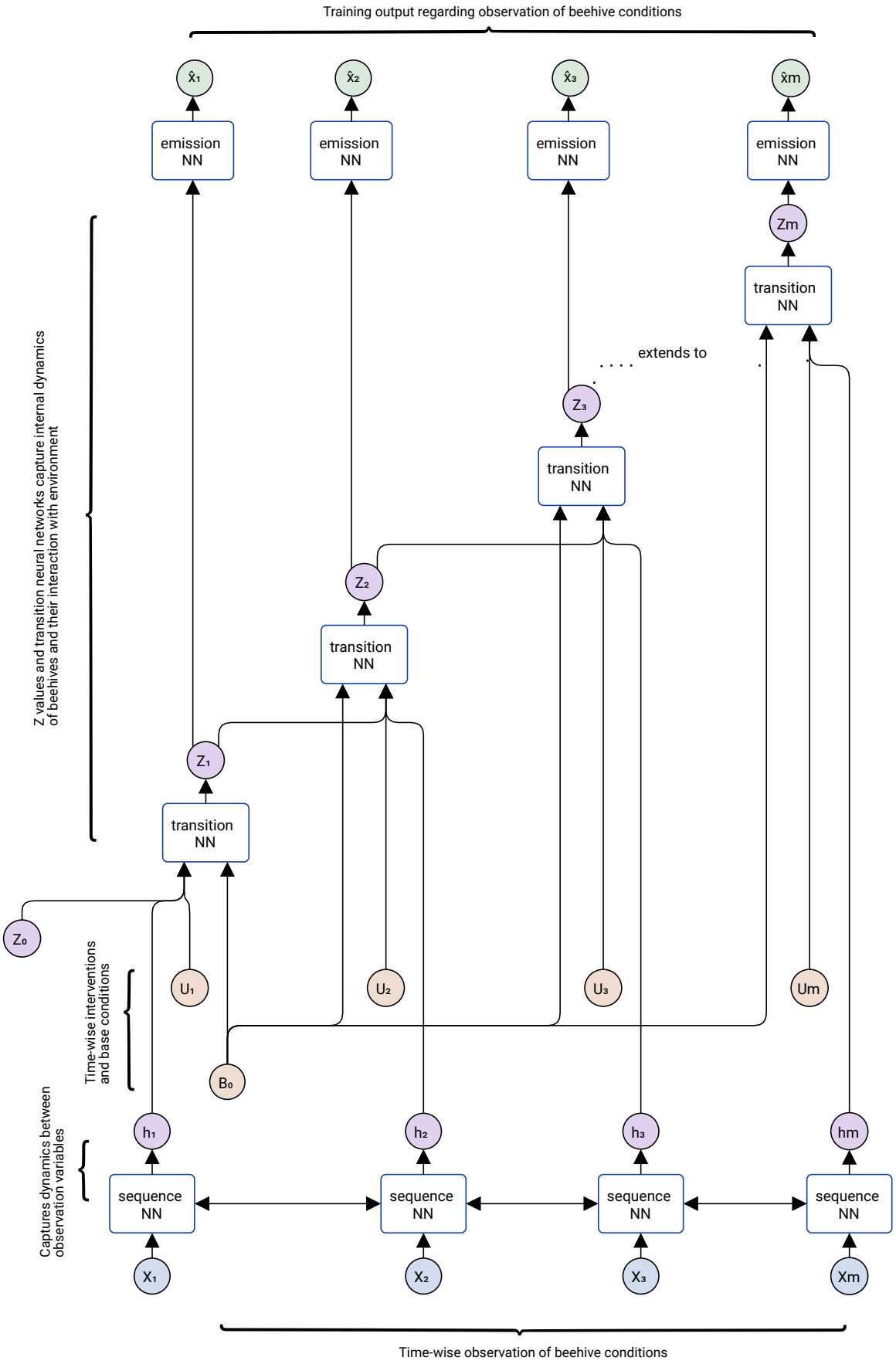


Figure 3. Training deep Markov model for beehive internal dynamics, conditioned on external factors.

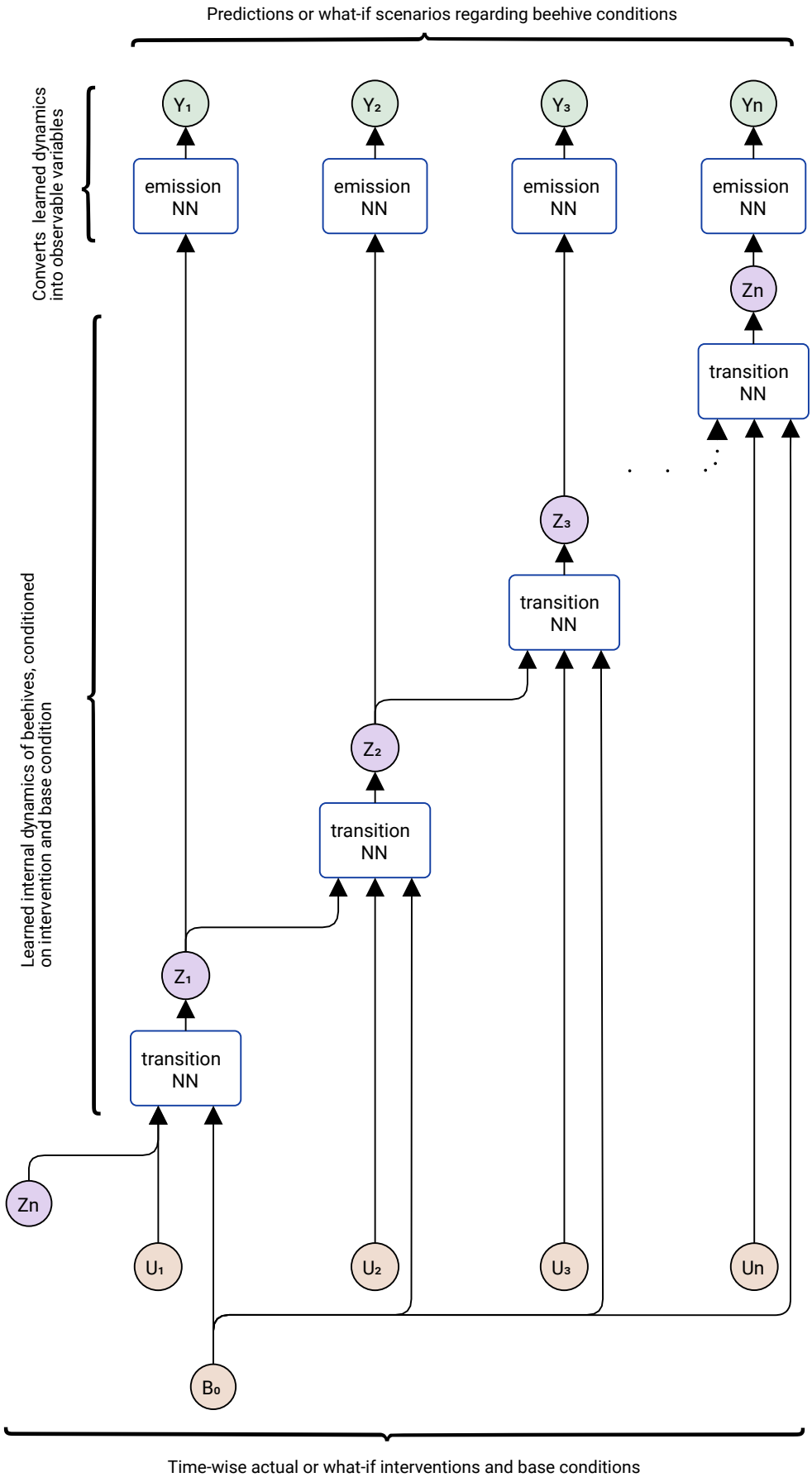


Figure 4. Prediction and what-if analysis of beehive conditions using trained deep Markov model.

3.4. Pretraining the Models for Population Dynamics and Mite Spread

A common challenge in applying neural networks to model Varroa spread, both within colonies (vertical) and between colonies (horizontal), is the large volume of training data required and the time needed to collect it from beehives. To address this challenge, we propose using existing mathematical and computational models to generate synthetic simulation data for the initial training phase. This strategy provides a well-informed baseline for the digital twin models, ensuring that they start close to realistic operating conditions. In turn, it can accelerate deployment and shorten the time needed to reach higher predictive accuracy.

There are several approaches for mathematically and computationally modeling the dynamics of vertical (within-colony) and horizontal (between-colony) spread of Varroa mites. To generate initial data for vertical (intra-colony) dynamics, we draw on the mathematical models of Torres and Torres [29] and Messan et al. [72], which we describe in the next section. An alternative that can be used in the same way is the agent-based model BEEHAVE [28,39,53], which simulates honeybee colony dynamics under Varroa pressure, alongside other stressors such as viruses, pesticides, and landscape change. The BEEHAVE model is a viable option to consider for future studies and for generating synthetic training data.

For horizontal (inter-colony) spread, we modeled infection transmission with an ERGM. In this formulation, colonies are nodes, and ties represent opportunities for Varroa transfer (e.g., drifting or robbing bees, shared foraging, equipment movement, or managed colony relocations). ERGMs specify the probability of these ties as a function of network structure (such as distance, clustering, or shared apiary) and colony attributes. The fitted model thus captures how spatial arrangement and management connectivity shape Varroa transmission between hives and enables the simulation of spread under alternative network configurations or interventions.

3.4.1. Vertical Spread Model: Within a Colony

Torres's model uses systems of differential equations to track daily bee populations and account for caste-specific survival rates [29]. Messan's study employs nonlinear delay differential equations (DDEs), incorporating explicit time delays and seasonal forcing, to analyze honeybee–mite population dynamics [72]. Either model can be combined with a horizontal spread component, for example, by casting the infectious states within a susceptible, exposed, infectious (SEI) or susceptible, exposed, infected, recovered (SEIR) framework for epidemiological invasion.

The population model for each colony was based on Torres' model (Equation (1)), where B_i represents the bee population at age i, S_i is the survival rate for that age group, and a_i is the rate at which hive bees become foragers.

$$\frac{dB_i}{dt} = \left(S_{i-1}B_{i-1} - B_i\right) a_i, \qquad i \geq 1 \tag{1}$$

The following equation describes the population dynamics of healthy hive bees, accounting for their life cycle, mite infestation, and recovery rates [29]:

$$\frac{dB_i}{dt} = \left(S_{i-1}B_{i-1} - B_i\right) a_i \; - \; \beta R(t) B_i \; + \; \gamma B_i^*, \qquad 21 \leq i \leq 41 \tag{2}$$

In this equation, $\beta R(t)Bi$ represents the transition from healthy to infested bees, with $R(t)$ indicating the proportion of infested bees, showing the infection's spread within the colony. For infested hive bees, the dynamics are given by

$$\frac{dB_i^*}{dt} = \left(S_{i-1}^* B_{i-1}^* - B_i^*\right) a_i + \beta R(t) B_i - \gamma B_i^*, \qquad 21 \leq i \leq 41 \tag{3}$$

This equation captures the effect of mite infestation on survival rates and the potential for recovery, with B_i^* indicating the number of infested bees and S_i^* their reduced survival rate. To represent the mite population, we used the following equation:

$$\frac{dM_i}{dt} = S_{M,i-1} M_{i-1} - M_i - D_i, \qquad 1 \leq i \leq 27 \tag{4}$$

Here, M_i is the mite population at age i, with S_{M_i-1} as their daily survival rate, which varies with the seasons. D_i accounts for mite deaths from grooming and the mortality of pupae in capped cells. The simulation began with an infected node at the network's centroid to study the impact of a central infection on the wider health of bee colonies. By adjusting parameters such as infection range (r) and infection rate (α), the model adapts to the changing landscape of Varroa mite transmission, providing valuable training data for training the digital twin models.

Existing research has already applied these equations, beginning with the original formulation by Torres and Torres [29] and extending to studies that explore different contexts and treatment schedules [26,73]. For example, Dasyam et al. [26] showed that varying treatment intensity and timing could markedly alter population trajectories, enabling recovery and sustained colony health when interventions are guided by what-if analysis.

3.4.2. Horizontal Spread Model: Among Colonies

Repeating the ERGM simulations with varying base infection and spread rates revealed several important patterns. As expected, the number of infected and high-risk colonies increases significantly when colonies are located in close proximity. Enhanced surveillance of high-risk nodes allows for the rapid identification and management of new infection epicenters. The model further demonstrates that early interventions can substantially slow the spread of infection. In addition, the results indicate the presence of critical thresholds beyond which containment measures become less effective, emphasizing the importance of timely detection and intervention. We anticipate that a digital twin system enhanced with predictive models would be capable of identifying these critical thresholds and epicenters, thereby making timely detection and intervention more feasible.

3.4.3. Pretraining the Models for Treatment Effects

In addition to modeling population dynamics and mite spread (Figure 5), it is also important to pretrain the models on the effects of treatments used to control Varroa infestations (Figure 6). Among the available strategies, hyperthermia, a heat-based method that reduces mite populations by exposing them to controlled elevated temperatures, provides a well-studied example [74]. While a single hyperthermia session does not eradicate mites completely, it significantly lowers their numbers and thus changes the trajectory of colony dynamics. By incorporating such treatment events into simulations, we extended the training data beyond natural progression and included realistic intervention scenarios. This approach allows digital twins to capture not only how infestations develop but also how different treatment strategies can alter outcomes. For example, simulations without intervention often lead to colony collapse within a relatively short period, whereas regular treatments can enable recovery and long-term colony survival (Figure 7). Such contrasting

scenarios enrich the training data with both successful and unsuccessful outcomes, which are crucial for robust model pretraining.

Finally, environmental conditions (e.g., seasonal cycles) play a major role in shaping bee population trajectories, with fluctuations driven by changes in birth rates and resource availability. Incorporating these cycles alongside intervention events makes the training data more representative of real-world conditions.

3.5. Dynamic Monitoring Strategy for Mitigating Disease Spread

Digital twin strategies must consider any constraints in sensing and communication resources while implementing effective and efficient treatment strategies. Thus, in our simulation, we considered a threshold of surveillance intensity needed for an effective treatment strategy. Our horizontal spread simulation highlighted the importance of specific parameters, such as the disease transmission rate and the spatial dynamics of the infection. Additionally, our model showed that infection rates and node susceptibility could differ greatly at various stages of an outbreak. As a result, a strategy of dynamic monitoring thresholds should be factored into digital twin design. This approach involves adjusting surveillance intensity in real time based on the observed patterns of disease spread.

To evaluate the effects of different infection intensities, we conducted several simulation runs, each time changing the infection rate value to represent various rates of disease spread. Through these simulations, one can monitor the emergence of high-risk nodes, which require increased surveillance to facilitate the early detection and treatment of new infestations. This adaptive response mechanism demonstrates the digital twin's ability to adjust to changing infection dynamics, where the diffusion of data from the sensor network updates the model and informs optimal interventions, such as hyperthermia treatment, based on the latest information.

3.6. Feasibility of Implementing Digital Twins

Our digital twin design employs a WSN for sensing, data exchange, and dispatching intervention commands. The network spans a large area and consists of nodes with heterogeneous resource levels. A typical node includes (i) a camera for Varroa mite detection, (ii) a communication module for uplinks and downlinks, (iii) temperature sensors to monitor hive and treatment conditions when hyperthermia is used, and (iv) a heating applicator to deliver hyperthermia. To ensure practicality, we quantified per-node energy consumption to assess the feasibility of continuous monitoring and on-demand treatment for a single hive, as node-level budgets ultimately determine system-level viability.

The power estimates in our feasibility analysis drew on measurements reported for widely used components. For the camera (e.g., ESP32-CAM (Ai-Thinker, Shenzhen, China)), the reported consumption is ∼30 mW in deep sleep and about 0.9–1.55 W during active image capture [75]. For the WSN uplink, an NB-IoT modem (e.g., Quectel BC66) typically draws ∼110 mA at 3.3 V during uplink (≈0.36 W), with idle/paging currents around 0.24 mA and power-saving-mode currents in the microamp range [76]. For temperature sensing, low-power digital sensors, such as the DHT22, draw roughly 1–2.5 mA while measuring and tens of microamps in standby [77]. Heaters used for Varroa control span a broad range, from low-power silicone pads (around 12 W) to vaporizer-class devices (near 150 W), depending on the treatment method [78]. These sources provide practical baselines for measuring monitoring and treatment energy in the proposed digital twin system.

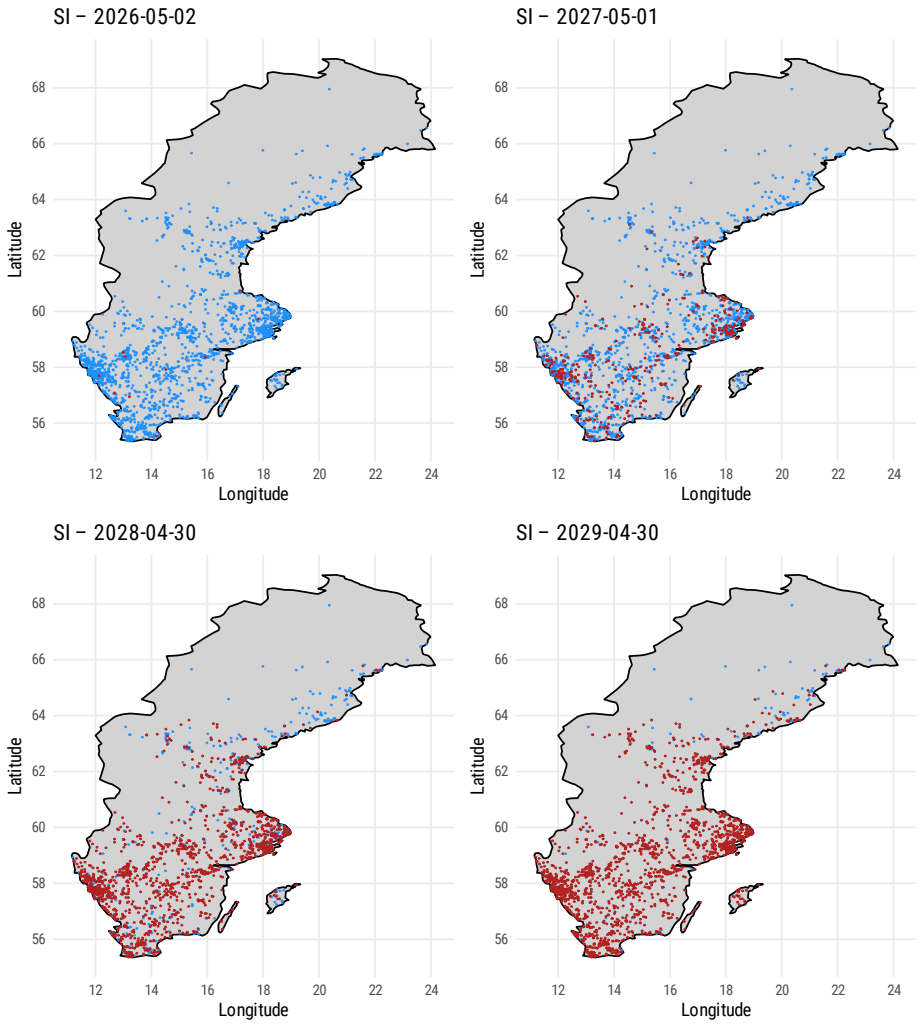


Figure 5. Varroa mite spread simulation: no intervention scenario. Infected colonies are shown in red; non-infected colonies are in blue.

Table 2 shows the energy consumption of various components during a single monitoring cycle. Using this data, we can estimate the annual energy requirements, considering both regular and increased monitoring schedules.

Table 2. Energy consumption per monitoring cycle.

Component	Operation	Power (W)	Duration (s)	Energy (W-s)
Camera Module (ESP32-CAM)	Capturing 5 images	0.90	20	18.00
DHT22 Sensor	Measuring temperature and humidity	0.0125	20	0.25
Microcontroller (ESP32)	Modem/light sleep	0.10	20	2.00
NB-IoT Modem (Quectel BC66)	Uplink transmission	0.36	5	1.80
NB-IoT Modem (Quectel BC66)	Idle/paging	0.0008	15	0.012
Total per Cycle				**22.06 W-s**

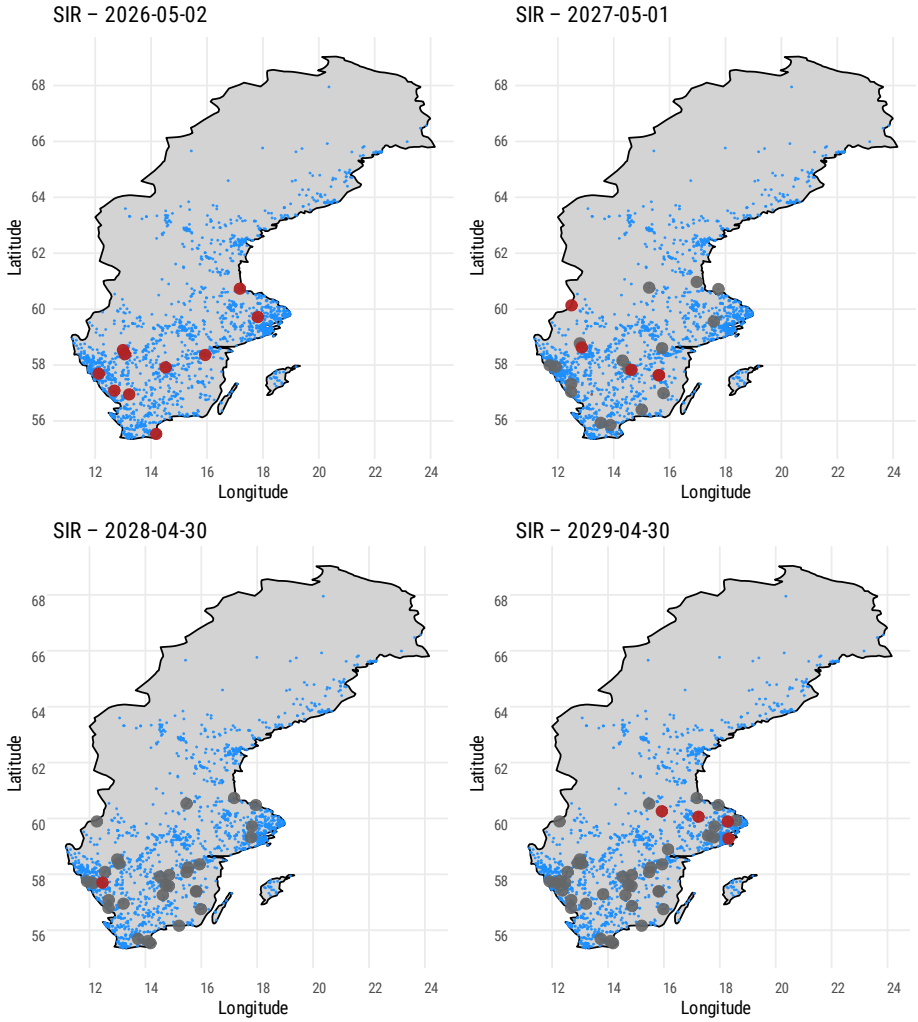


Figure 6. Varroa mite spread simulation: periodic treatments scenario. Infected colonies (red) and recovered ones (gray) are enlarged for the sake of visibility.

There is also considerable variation in treatment duration, ranging from 90 min at 42 °C with the Bee Ethic system [79] to about 150 min (20 min ramp + 130 min hold at 42.5 °C) in the Vatorex in-comb heating protocol [78], and up to 2.5 h per cycle in other approaches [80]. In all cases, the treatment elevates the capped brood temperature from its normal range of 34–35.5 °C to approximately 42–42.5 °C, an increase of about 7–8 °C. Seasonal differences in heat leakage rates further influence the actual energy required to maintain these elevated temperatures.

The actual power required for a full hyperthermia session is difficult to determine, as it depends heavily on the heating system design, whether it heats the entire hive at once or operates frame by frame, and ambient temperature. For practical purposes, we assume that access to grid power is limited and that heating must instead be supplied by rechargeable batteries, which can be recharged via photovoltaic panels or supported by portable generators. Only limited information is available on the actual power consumption of commercial devices or experimental prototypes. For instance, a patent for a frame-heating device reports a demand of 400–450 W [81]. Similarly, Porporato et al. [79] describe a system powered by a 24 V supply (can be equipped with a solar panel and a battery), requiring operation every 25 days for 90 min per frame. Assuming a foil heater of 25 W per frame, a 10-frame hive plus a 20% margin results in a total of about 450 W, consistent with the patent report. Thus, 450 W can be considered a reasonable median estimate for one full

treatment cycle. It should be noted that with this power requirement, 7–8 treatment cycles per year would each necessitate a battery replacement if no recharging source is available or if investing in a higher-capacity battery is not feasible.

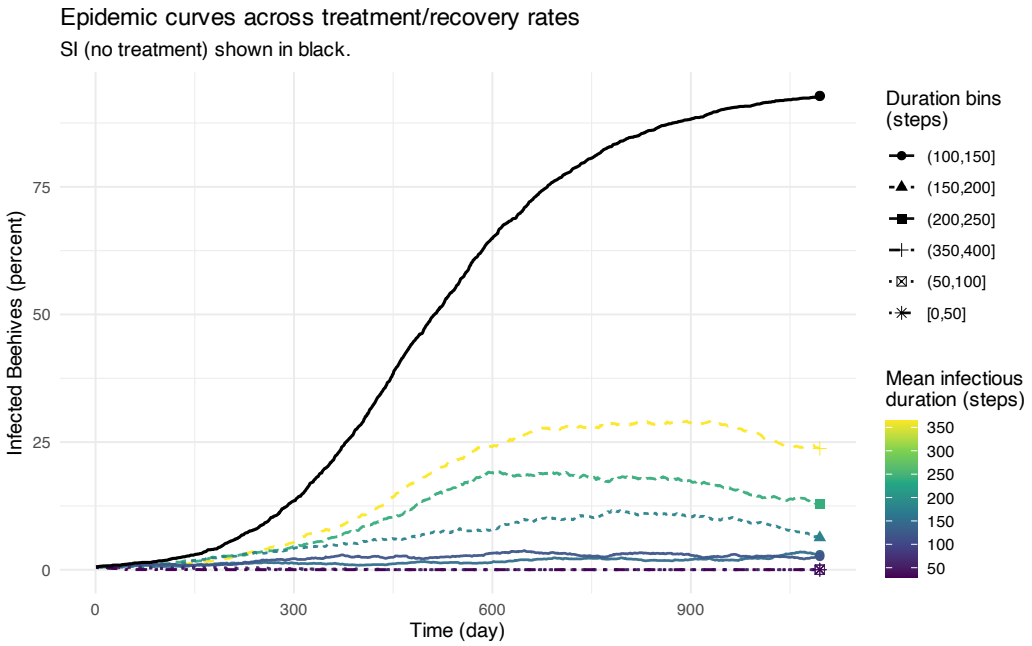


Figure 7. Varroa mite control through treatment scheduling. In the absence of intervention, within-colony dynamics are well approximated by an SI process, and infection tends toward widespread prevalence. When treatments (e.g., hyperthermia) are applied according to a schedule, the system resembles an SIR process, allowing prevalence to be maintained below target thresholds and preventing system-wide takeover.

Table 3 calculates the high-end and low-end energy consumption for a treatment cycle with 10 frames (assuming 2.5 h for each), including both heater and microcontroller operations. While a detailed economic analysis is beyond the scope of this paper, these figures highlight that the cost and complexity of storing and supplying the required energy may represent a significant barrier to large-scale adoption and ease of use of such systems.

Table 3. Energy consumption for a single treatment cycle (2.5 h heating).

Component	Operation	Power (W)	Duration (s)	Energy (W-h)
Microcontroller (ESP32)	Temperature regulation (active)	0.10	9000	0.25
Heater (12 W low-power pad) **Total per cycle (10 frames)**	Hyperthermia heating	12	9000	30.00 **300.25 W-h**
Heater (65 W medium pad) **Total per cycle (10 frames)**	Hyperthermia heating	65	9000	162.25 **1620.25 W-h**

It is important to note that these energy consumption calculations are based on typical values, considering certain fixed parameters specific to regions and devices. An advantage of digital twins is their ability to provide more accurate and factual calculations and predictions of these consumption patterns by incorporating variable environmental and device factors.

4. Discussion

The digital twin design illustrated in Figures 1 and 2 demonstrates how sensor-enabled beehives, predictive and generative models, and treatment actuators can be integrated into a

closed-loop system. Within the DRM framework [43], this constitutes the Prescriptive Study stage, offering a structured concept that can subsequently be refined through descriptive evaluation and be implemented later. The looped workflow highlights how real-time data collection, model updating, and treatment interventions can converge into an adaptive cycle, where interventions are continuously assessed against outcomes [32,33,35].

An important implication of this architecture is its ability to support generative scenario modeling [82], where multiple potential responses are envisioned and compared before action is taken. This feature is crucial in apiculture in general and in controlling Varroa mites in specific, where environmental variability, management constraints, and policy considerations must all be accounted for [2]. By representing interventions as scenarios with different cost, feasibility, and impact profiles (from the beekeeper level to the national level), the digital twin allows for decision-making that extends beyond technical optimization to include socio-ecological dimensions [35].

The modular nature; the spanning sensing, modeling, and scenario generation; and actuation of the design also support scalability. For example, this allows for deployment across different hive densities, landscapes, or management regimes while retaining the same underlying framework.

The reviewed approaches to mite detection underscore the central role of mite sensing (visual or non-visual) in enabling a digital twin for Varroa management. The performance reported for ViT-based systems, such as VIT4V [20], indicates that state-of-the-art accuracy is attainable even under realistic hive conditions, where blur, occlusion, or environmental noise are present. This level of robustness is essential for supporting a reliable digital twin, as predictions and scenario models depend directly on detection quality. By incorporating such models, the twin is not only able to monitor mite levels more effectively but also to feed validated infection data into downstream forecasting and scenario generation.

At the same time, the landscape of visual detection methods is evolving rapidly. The integration of model-agnostic design choices within a digital twin allows for flexibility to adopt new detectors as they emerge, ensuring that the system remains adaptable to hardware constraints and advances in edge-efficient architectures (i.e., in-beehive sensor and computation capacities).

The integration of vertical and horizontal spread models into a digital twin highlights the value of combining established epidemiological approaches with emerging generative time-series models. Traditional mathematical and computational models (such as ERGMs [30]) offer interpretability and clear parameterization, while generative time-series models extend this capacity by supporting counterfactual simulations, uncovering hidden dynamics, or even hinting about missing variables in the picture. Although the output of generative time-series models is just another set of numerical time series, it is still not very hard to elicit stories and scenarios from it and communicate it effectively to stake-holders and decision-makers.

Another implication is that probabilistic state-space models, such as DMM [63], provide a natural way of representing uncertainty in colony dynamics. By quantifying uncertainty and enabling counterfactual projections, these models allow the digital twin to evaluate intervention strategies under varying assumptions and to anticipate outcomes beyond those directly observed [63]. This is important in apiculture, particularly in the control of Varroa, where variability in environment, management, and policy creates complex drivers of Varroa spread dynamics [1] that cannot be fully captured by deterministic models.

At the same time, the iterative updating loop ensures that models remain adaptive to changing colony, environmental, and managerial conditions, mitigating the effects of data drift and probably even providing hints about possible concept drift. This adaptive

feedback, embedded in the digital twin [31], represents a shift from static models toward continuously learning systems that evolve alongside the colonies they represent. Such adaptability is essential if the twin is to remain reliable across diverse environmental conditions and management regimes.

The pretraining strategy in our digital twin design highlights a key advantage: the ability to leverage established mathematical and computational models to accelerate the development of learning-based systems. By generating synthetic datasets from differential equations [29] or agent-based models [28], digital twin components can begin from a well-informed baseline rather than an uninformed state. This approach is particularly valuable in apiculture, where collecting sufficiently large and diverse field datasets is time-intensive and logistically difficult. It can also be more persuasive regarding a beekeeping community participating in a research project when they can see that the system can be informative from the first day.

The ability to identify and prioritize high-risk nodes demonstrates how real-time feedback from the sensor network can be converted into adaptive responses. In practice, this means that not every hive needs to be monitored and equipped with treatment actuators with the same intensity at all times; rather, monitoring and treatment can be scaled in proportion to predicted risk, enabling more efficient use of resources.

The feasibility analysis highlights how energy demand becomes the central design constraint when implementing digital twins in beekeeping, specifically in the management of Varroa mites. While sensing and communication draw modest power, treatment operations, particularly hyperthermia, dominate overall consumption. In practice, this suggests a roadmap for deployment: starting with monitoring-only digital twins that are only equipped with sensor packages to build data streams. Enhancing beehives using a treatment applicator can be carried out in smaller amounts or moved around based on predicted need.

Last but not least, while this conceptual design was developed for controlling Varroa mites in beehives, its structure can be directly adapted to other precision agriculture applications. The combination of sensing, generative scenario modeling, and targeted interventions is equally relevant for managing crop pests, monitoring plant health, or supporting livestock care, where similar constraints on energy use, data availability, and timely decision-making apply.

5. Conclusions

The digital twin system's conceptual design presented in this paper demonstrates how adaptive monitoring and treatment strategies can be achieved through a continuous cycle of sensing, model updating, exploration of what-if scenarios, and interventions. By integrating real-time data with predictive and counterfactual models, the system dynamically adapts to changing conditions in mite spread, enabling more effective and timely responses. Additionally, our approach of synthetically generating training data for a range of population, spread, and treatment scenarios enables pretraining of the digital twin system. This strategy facilitates smoother deployment and accelerates the achievement of effective results.

The simulated scenarios further underscore the importance of treatment timing in mitigating disease impact, highlighting the need for a system capable of adapting its actions based on real-time conditions and predictions. This aligns closely with the concept of digital twins, where the physical state of bee colonies is continuously mirrored in a virtual model to determine the most effective intervention strategies. Evaluations of mite detection and energy consumption, together with simulations of Varroa mite control, demonstrate the functional feasibility of our proposed digital twin system design.

Further investigation is needed to evaluate different generative models for time series with respect to their capacity for modeling long-horizon time series, accommodating diverse data characteristics, and generating both factual predictions and counterfactual scenarios for what-if analysis in the context of Varroa mite control. The overall system will also require a WSN capable of monitoring a large number of beehives across wide areas, ensuring accurate inputs for effective mite management strategies. In addition, further work is necessary on visualization and user-friendly interaction design, as the digital twin system is ultimately intended to support beekeepers' decision-making for timely interventions. Scaling and real-time testing remain critical areas for future exploration to fully realize the potential of digital twins in Varroa mite control.

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